Revolutionizing drug R&D with Lab-in-a-Loop

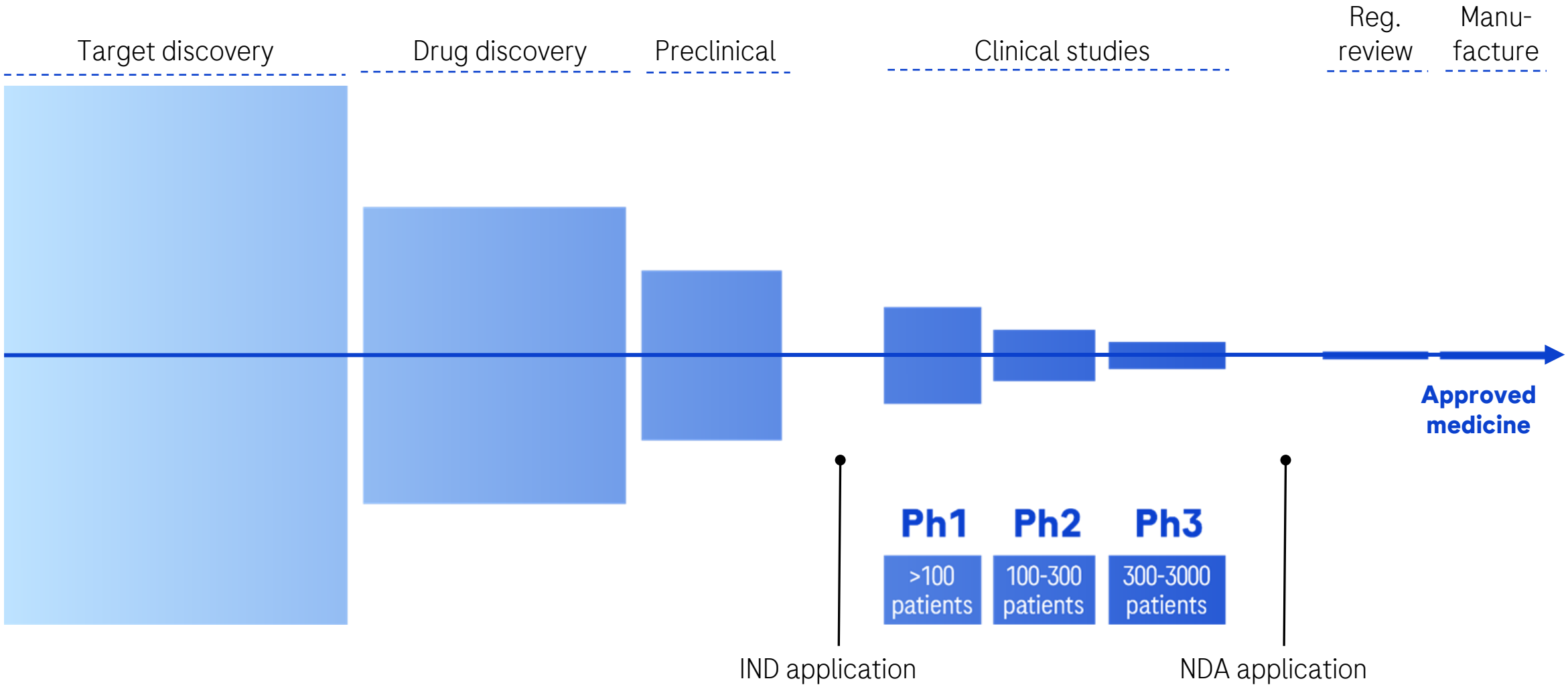
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Genentech Research and Early Development (gRED)

Genentech
A Member of the Roche Group

September 28th, 2026

Drug R&D is a challenging process with a low success rate

Typically, only ~1-2 of 100 programs initiated in research will lead to a launch



Why is drug R&D so difficult?

BIG NUMBERS

Enormous search spaces

MULTI-SCALE

Atoms, genes, proteins, cells, tissues, organs, patients

MULTI-MODAL

Multiple measurements / views of each entity

VARIED

No two research paths are quite the same

AI has the potential to tackle each of these challenges

BIG NUMBERS

Enormous search spaces

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Atoms, genes, proteins, cells, tissues, organs, patients

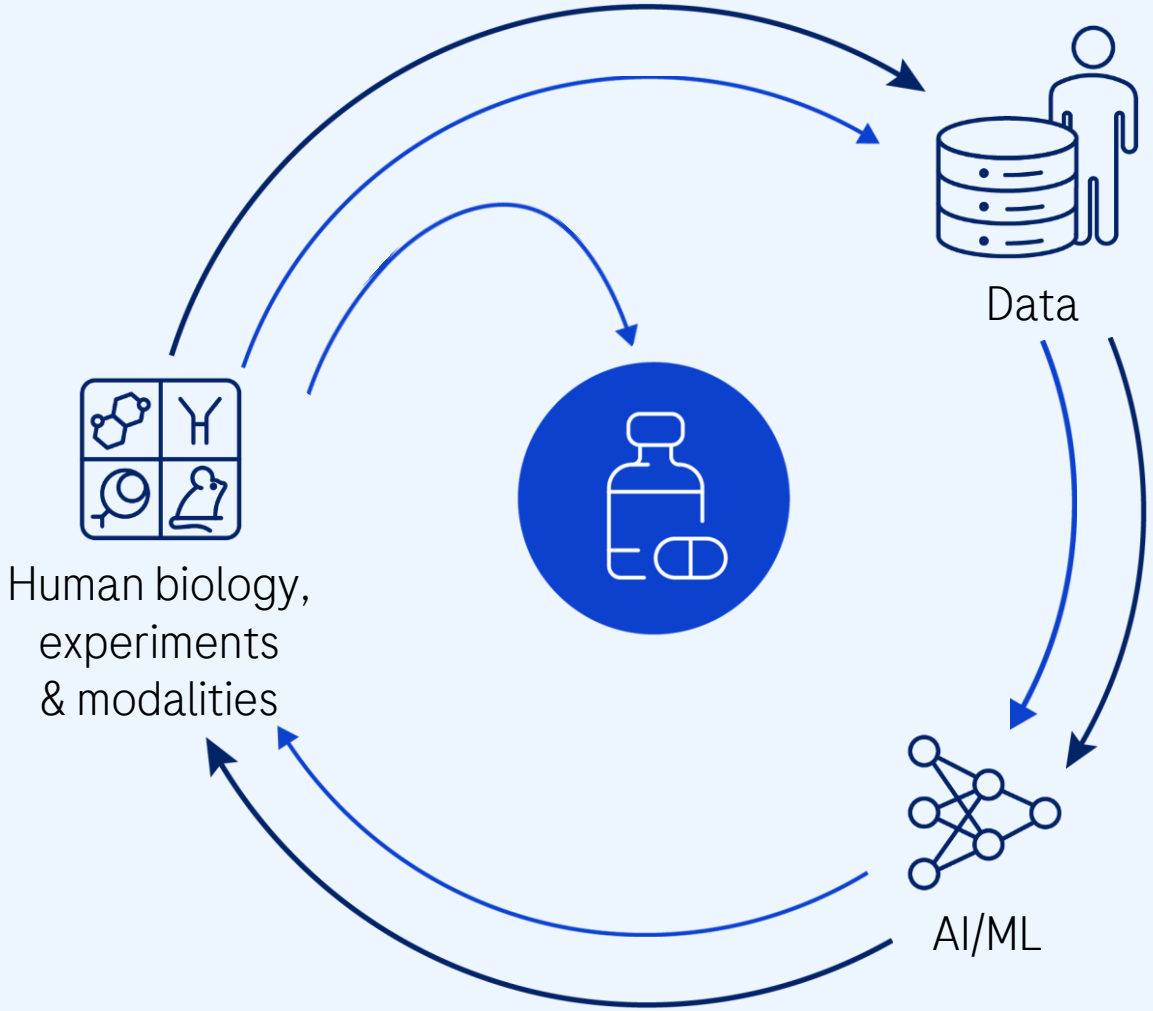


MULTI-MODAL

Multiple measurements / views of each entity

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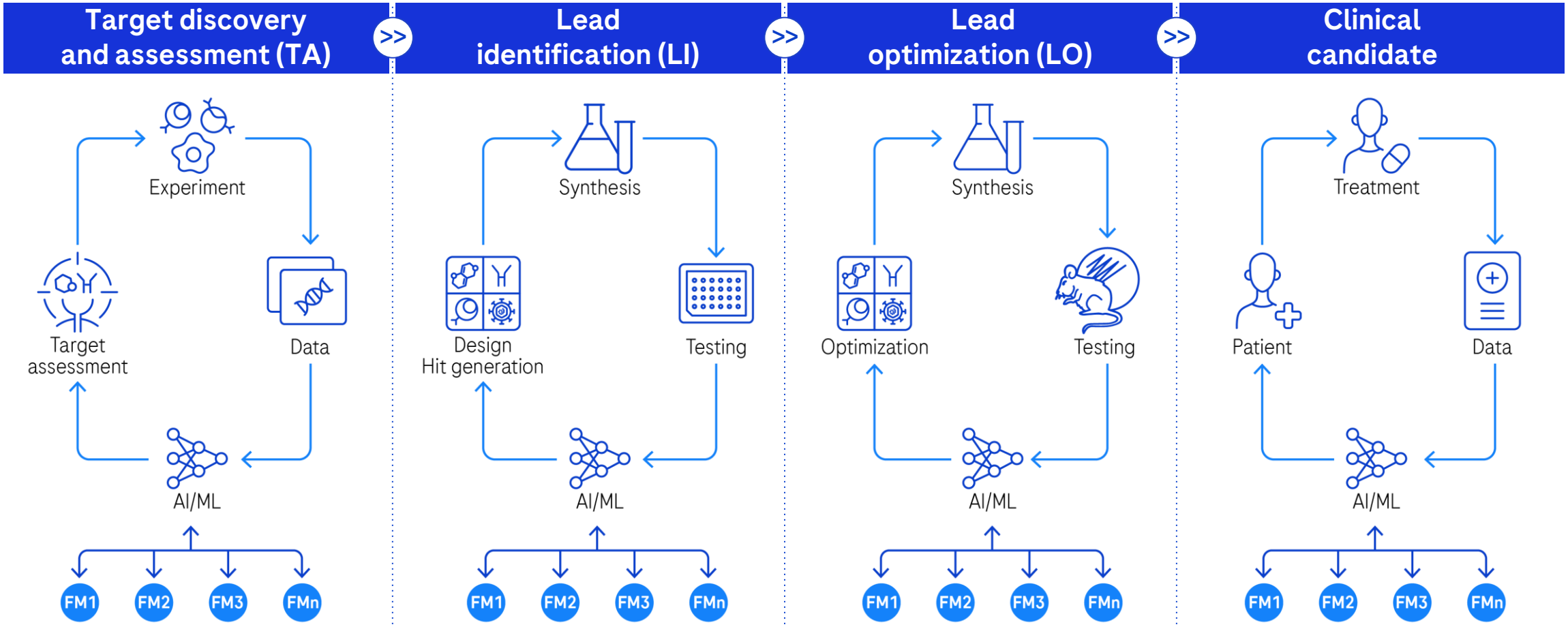
No two research paths are quite the same



LAB
(OR CLINIC)
IN THE LOOP

Lab-in-the-Loop in drug R&D

Iterations with AI embedded E2E, across each component, from target discovery to the clinic



AI/ML, Artificial Intelligence and Machine Learning; FM, Foundation model; LI, Lead identification; LO, Lead optimization; R&D, Research and development; TA, Target assessment

Our journey from build to delivery



Our journey from build to delivery

Strategy-driven investment across talent, AI capabilities, org design, data generation capacity & compute

2020 2021 2022 2023 2024 2025 2026

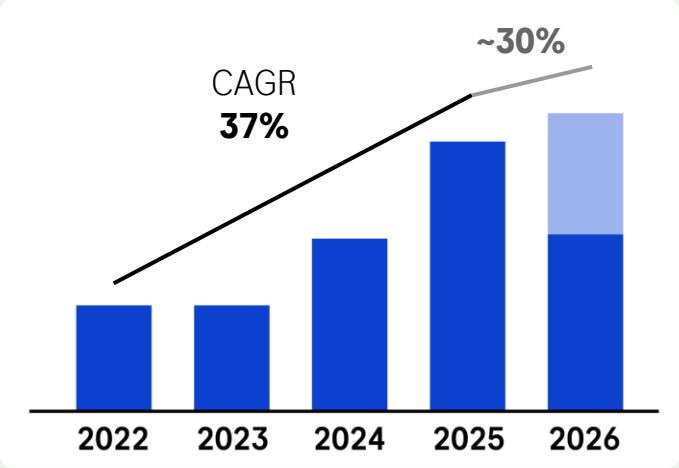
	2020–2022 **Build the foundation**	2023–2024 **Scale…**	2025–2026 **Transform…**
Strategy & capability	• Lab-in-the-Loop strategy • Multiplicative Levers	• Prescient expansion (all molecules) • Herophilus[1] automation software • AWS Research Agent collaboration	• Medra physical AI, MHS (Anthropic[1]) • Autonomous Lab Accelerator
Organization & talent	• BRAID (biology AI) • PrescientDesign[1] (antibodies AI)	• gRED Computational Sciences (gCS) consolidated	• Computational Sciences Center of Excellence (serves both REDs)
Data generation	• Cell & Tissue Genomics; Translational Genomics • Recursion collaboration[1]	• A-alpha Bio collaboration[1]	• Foundational Datasets for FMs initiative • Caris collaboration • LitL2.0 initiative
Compute & software		• NVIDIA partnership[1]	• OpenAI pilot partnership[1] • AI factory (NVIDIA)
Example impact milestones	1st LitL small molecule cycle 1st LitL large molecule cycle	1st LLM trained on Roche data 1st IBD pathobiology map + 15 LI targets 1st LO small molecule scaffold with AI-supported design	1st LI target enters with AI agent support 1st Antibody with AI support enters development portfolio (FPI) AI participating: 100% large molecules / 50% small molecules[2]

1. Partnering/acquisition; 2. ~90% of eligible projects | Notes: BRAID: Biology Research AI Department; AI: artificial intelligence; LLM: large language model; LitL: Lab in the loop; LI: lead initiation; LO: lead optimization; SM: small molecules; IBD: Inflammatory bowel disease; gCS: gRED computational sciences;

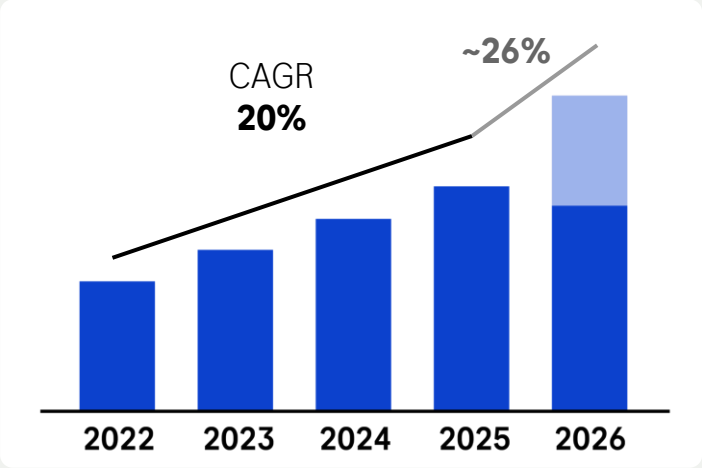
Steady increase in research portfolio productivity 2022-2026

Increase from LI-Go to LO-Go to development candidates under nominally flat resources and halving LI CT

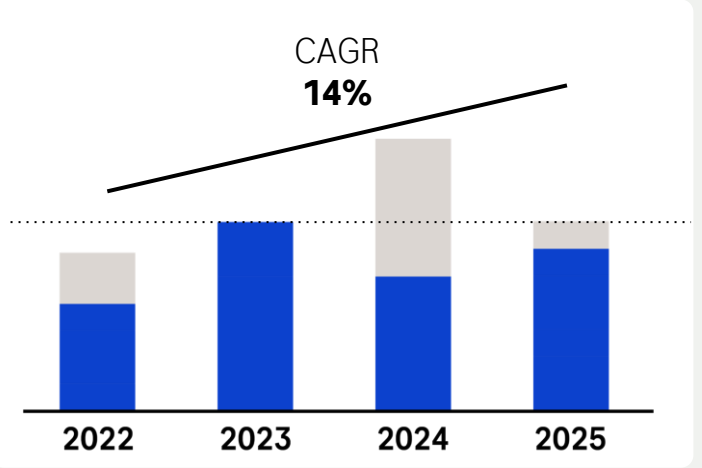
Lead Identification Go



Lead Optimization Go



Internal Development Go



2022-2025 growth

+155% LI Go

+75% LO Go

+50% internal DevGo

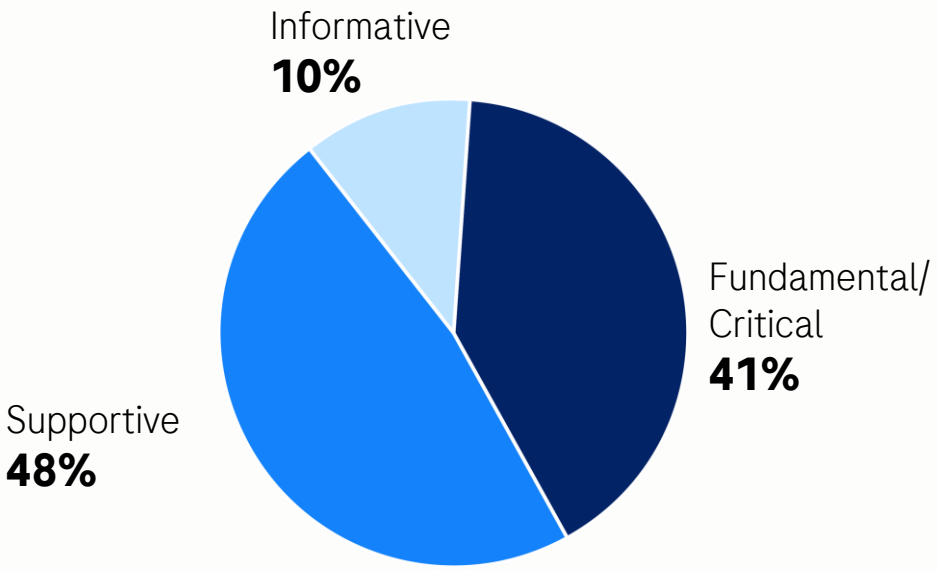
Actual · Forecasted · Business deals · ····· Historical number

CAGR, Compound annual growth rate; LI, Lead identification; LO, Lead optimization

AI + computation contribute across the pipeline

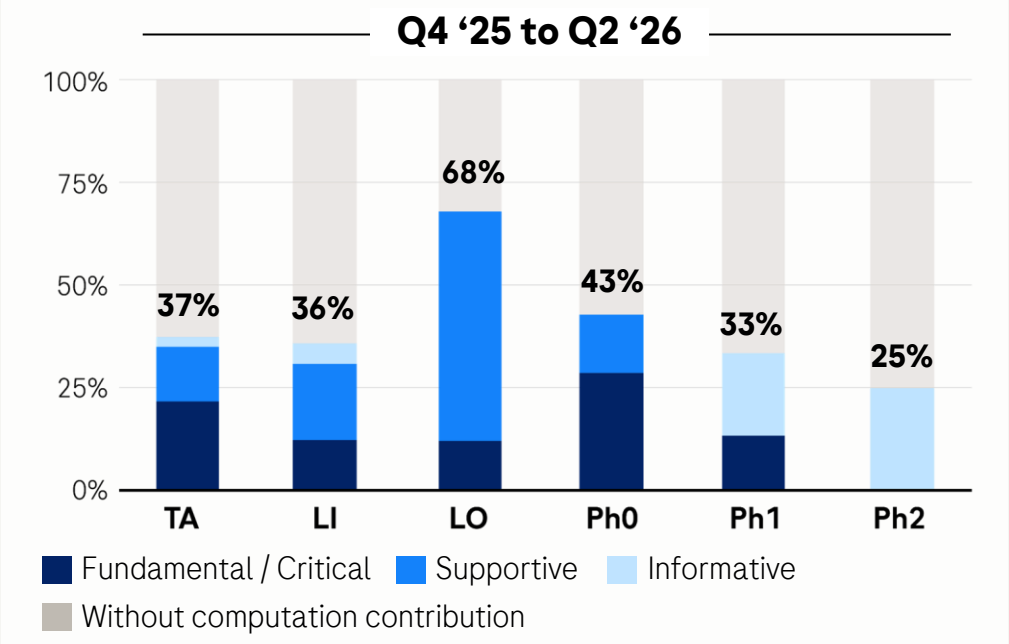
40% of Q4 ’25 through Q2 ’26 pipeline decisions had a tracked AI / computational contribution

Role of contributions in decision making



- **Critical:** Materially influenced
- **Supportive:** Meaningfully impacted decision confidence
- **Informative:** Informed thinking

Impact on pipeline decisions by phase

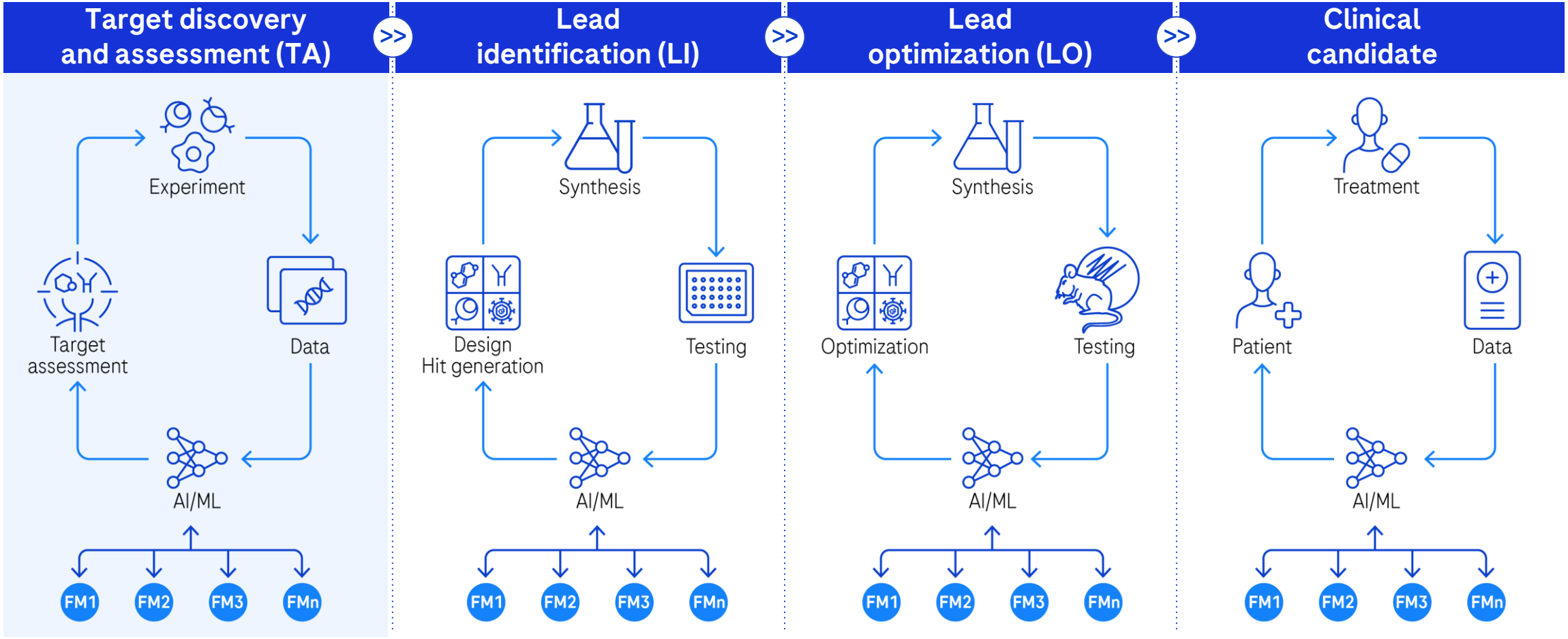


Cumulative coverage: 40%
(Q2 coverage reached 44%)

AI, Artificial intelligence; CS CoE, Computational sciences center of excellence; LI, Lead identification; LO, Lead optimization; Ph, Phase; TA, Target assessment

Lab-in-the-Loop in drug R&D

Iterations with AI embedded E2E, across each component, from target discovery to the clinic

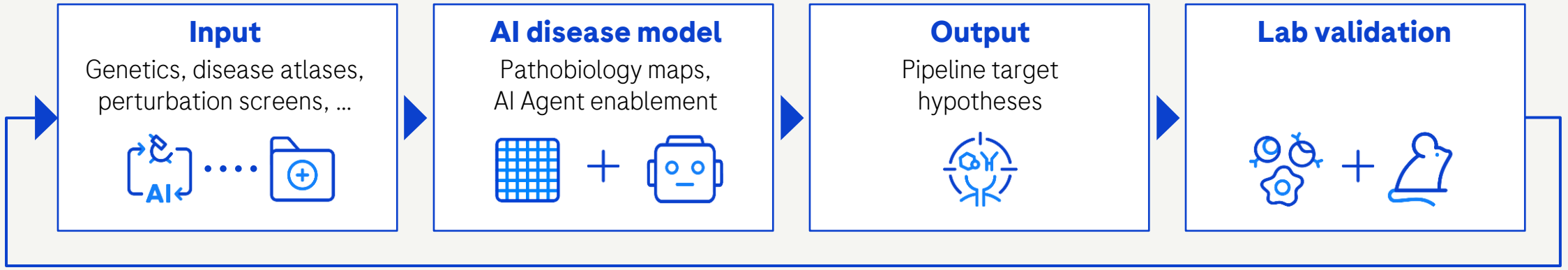


AI/ML, Artificial Intelligence and Machine Learning; FM, Foundation model; LI, Lead identification; LO, Lead optimization; R&D, Research and development; TA, Target assessment

Increase in high quality volume of research portfolio entries

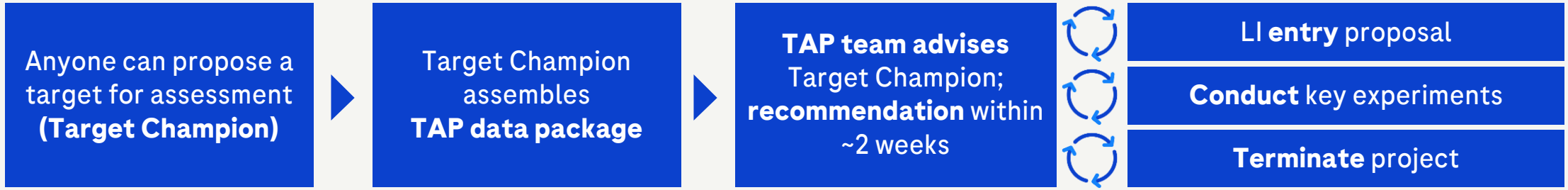
Large-scale human and lab data, AI, and new processes increase high-quality volume and speed

Innovative scientific foundation: AI-enhanced pathobiology maps for target discovery



New enabling process: Target Assessment Portfolio (TAP) Teams

Cross-functional system to ensure speedy target assessment, expert guidance and prioritized resourcing

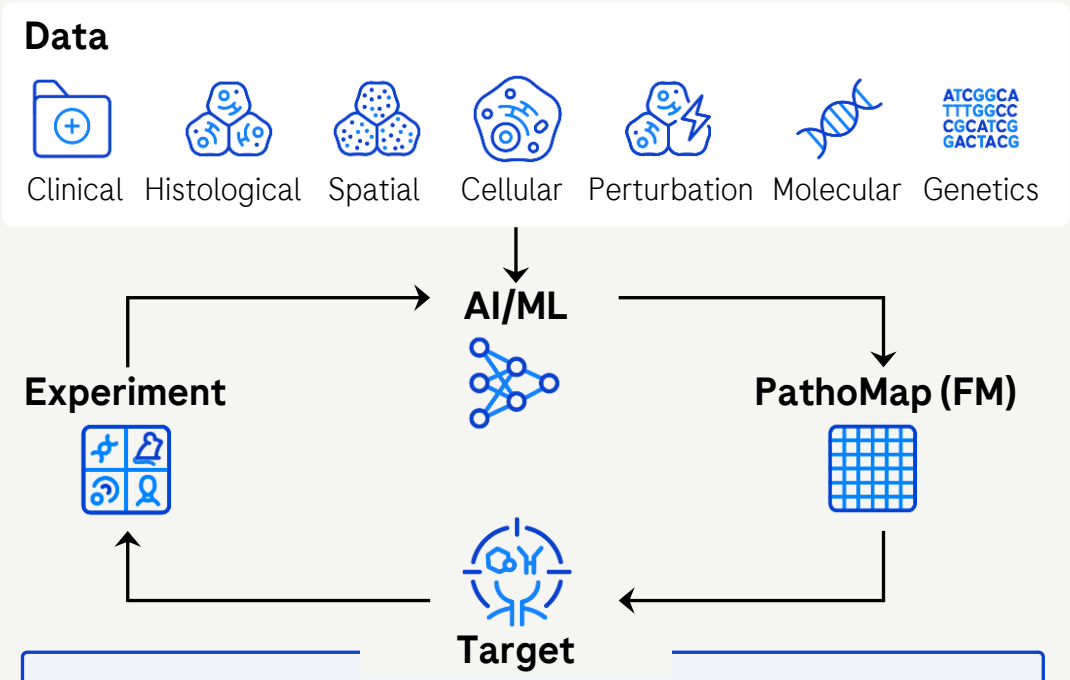


AI, Artificial intelligence; LI, Lead identification; TAP, Target assessment portfolio

Pathobiology maps have scaled and accelerated our pipeline

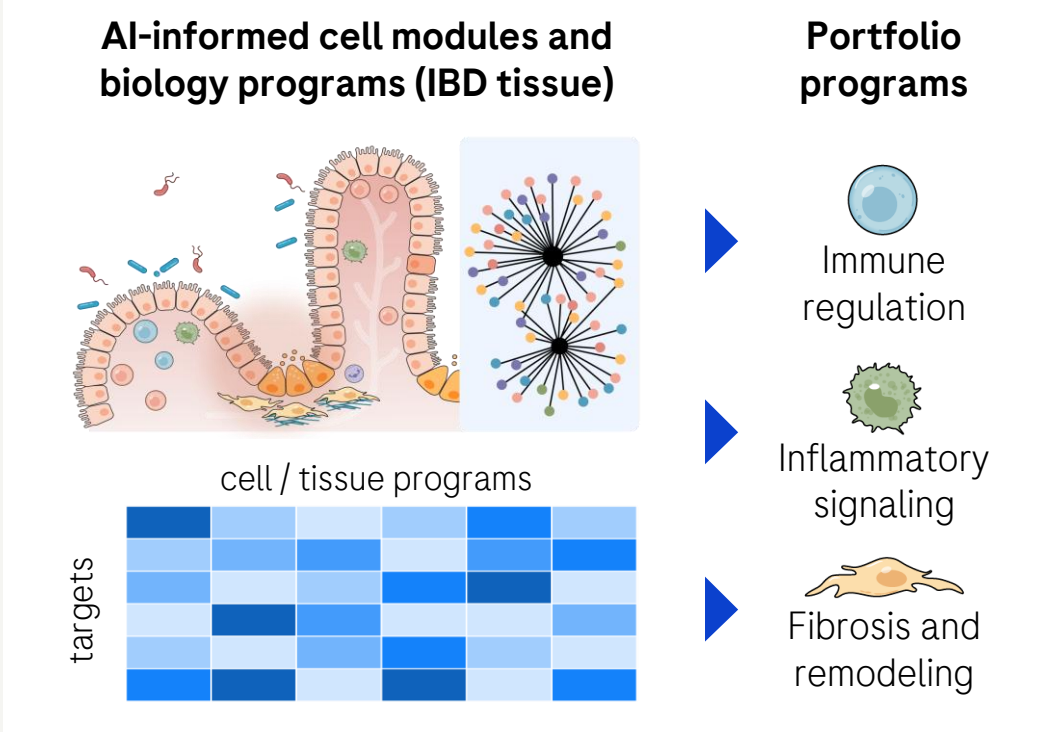
Example: IBD pathobiology maps

Goal: Data & FMs to help break the efficacy ceiling



- New drug targets
- Molecule discovery
- New biomarkers
- New combination therapies

Example: IBD Pathobiology Map v 2.0



- Example map: cells x spatial context x human genetics
- Multi-modal data power IBD target discovery

AI, Artificial intelligence; IBD, Inflammatory bowel disease; FM, Foundation model; PathoMap, Patho-biology map

AI-enhanced pathobiology maps across our TAs

Oncology ✓	Immunology ✓	Neurology ✓	Ophthalmology ✓
Cell surface targets across tumors	COPD IBD	Alzheimer's MS Parkinson's ALS	AMD

1 YEAR

First map deployed (IBD)

6 WEEKS

4 neurology maps deployed

+155%

increase in LI entries in 2 years

ALS, Amyotrophic lateral sclerosis; AMD, Age-related macular degeneration; COPD, Chronic obstructive pulmonary disease; IBD, Inflammatory bowel disease; LI, Lead identification; MS, Multiple sclerosis

Foundational Datasets for Foundation Models initiative

Large-scale data generation dedicated for training of AI models of relevant disease biology

Foundational datasets (FD)

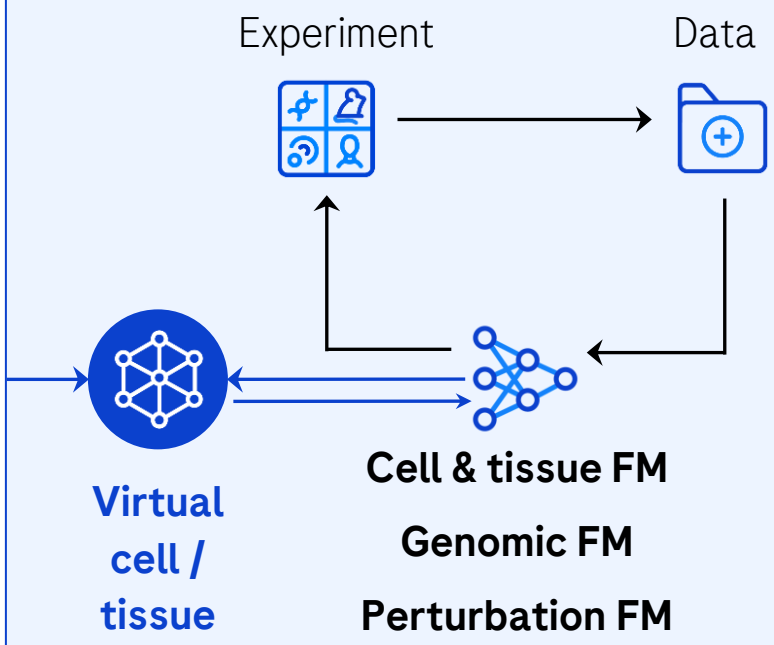
Sequence-to-phenotype
Human genetics
Massive scale lab assays

Perturbations
Genetics and compounds
Cells, organoids, tissues

Patient tissues
Healthy & disease

Clinico-genomics
Mechanisms & patient outcomes

Foundation models (FM)



Target discovery and prediction
→ Find causal drivers

Mechanistic insight
→ Understand how targets alter cell or tissue state

Target-hopping
→ Find alternative therapeutic targets

Target deconvolution
→ Identify what drives an observed compound's effect

Safety
→ Detect on- and of-target effects

FD, Foundational datasets; FM, Foundation models

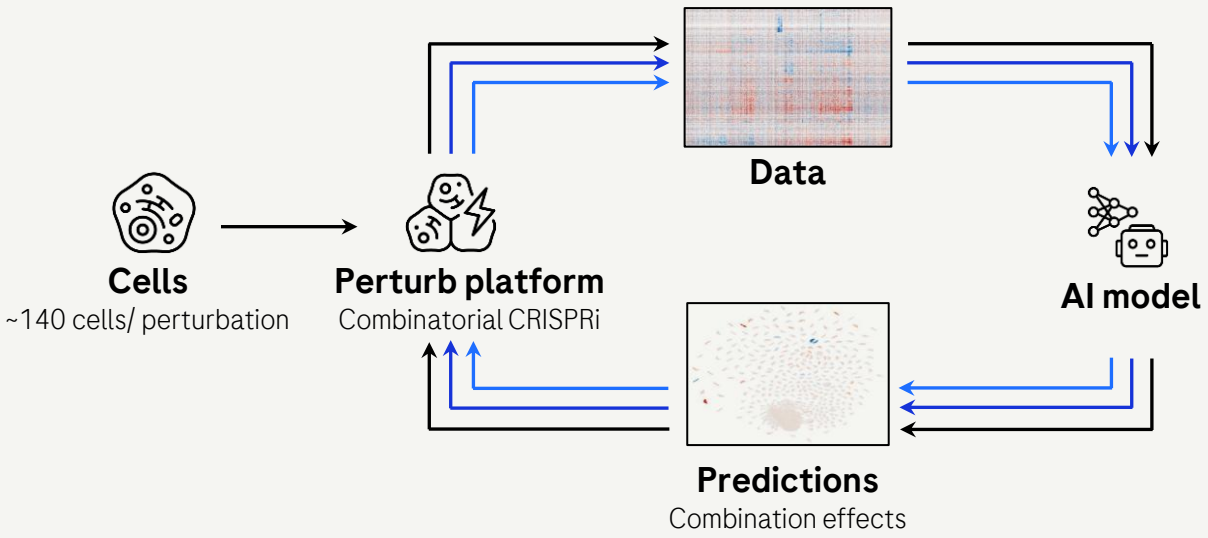
Tackling biological search spaces with active learning

Iterative AI-driven sampling efficiently explores the search space

Example: Iterative Perturb-Seq loop

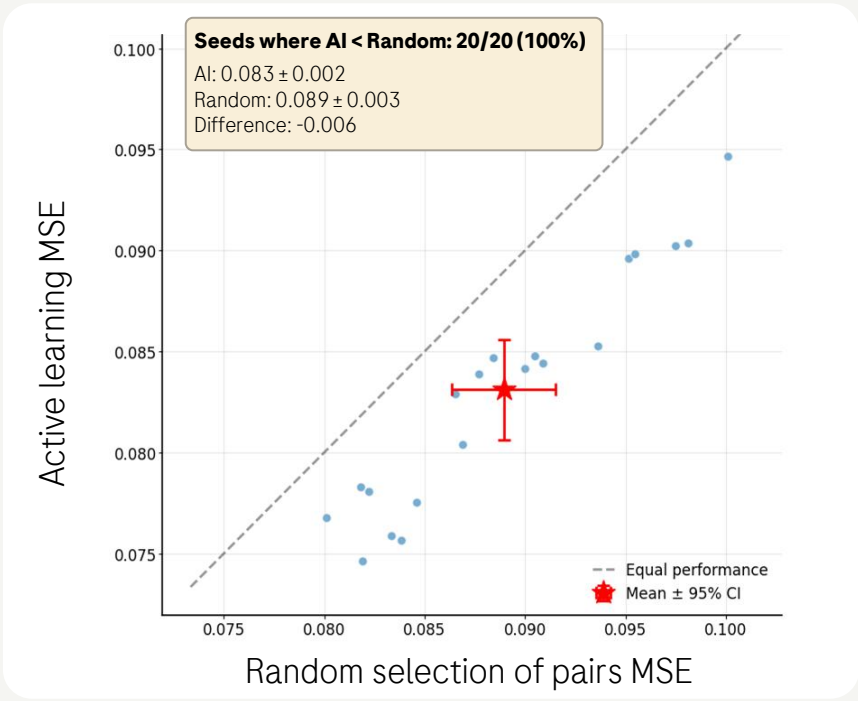
Search space: ~180k pairs over 600 impactful genes

Sampling: ~600 pairs / round



Round #1	Round #2	Round #3
No AI **~700** selected pairs	AI selected **600** pairs +Baseline: **400** randomly selected pairs	AI selected **600** pairs +Baseline: **400** randomly selected pairs

Active learning *vs.* baseline selection



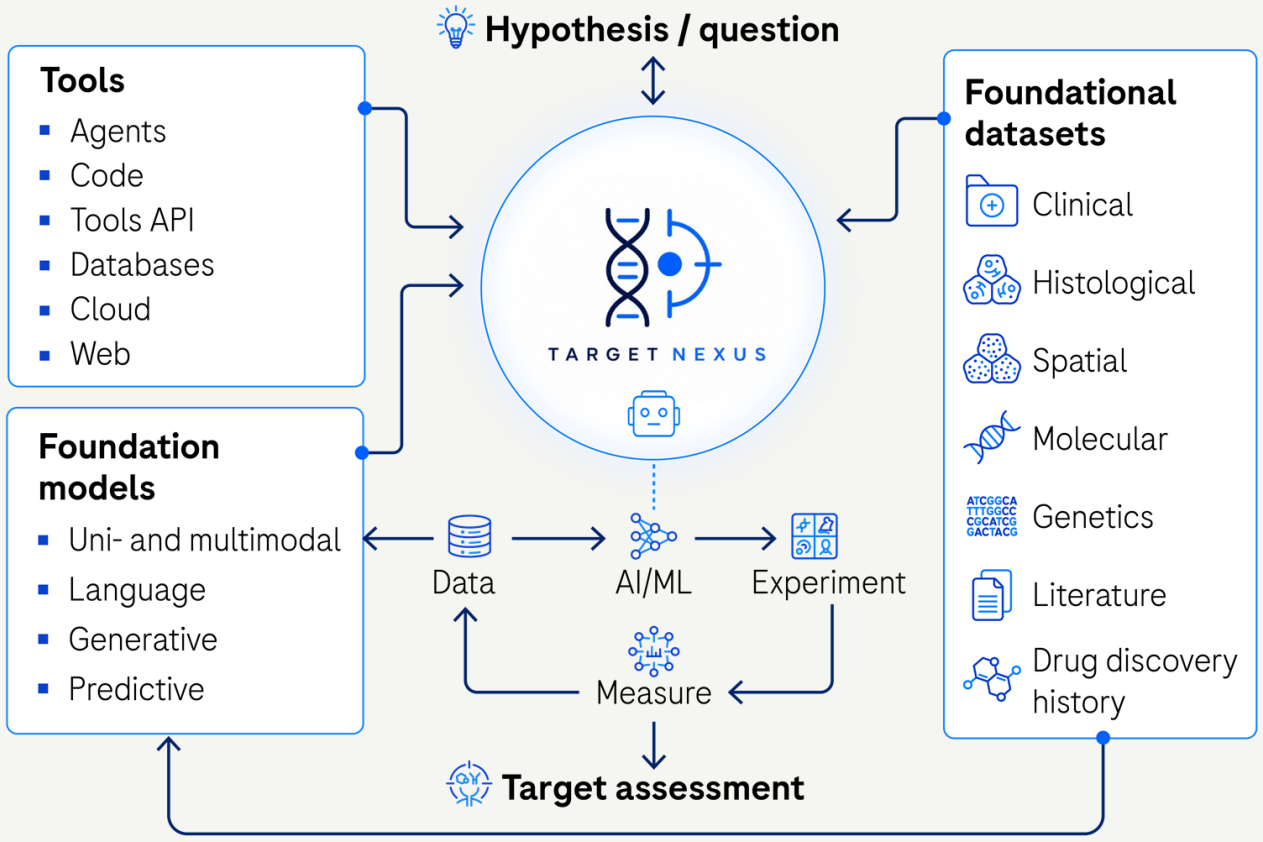
Active learning delivers better predicting model than strong baseline selection (random pairs over 600 impactful genes)

AI, Artificial intelligence; k, Thousands; MSE, Mean squared error

TargetNexus

Our AI agent for target assessment on track to be part of 80% of research portfolio decisions by Q4

Target discovery and assessment



Coverage across the pipeline

Target Biology

- Causal mechanisms
- Disease biology
- Disease models

Drug Discovery

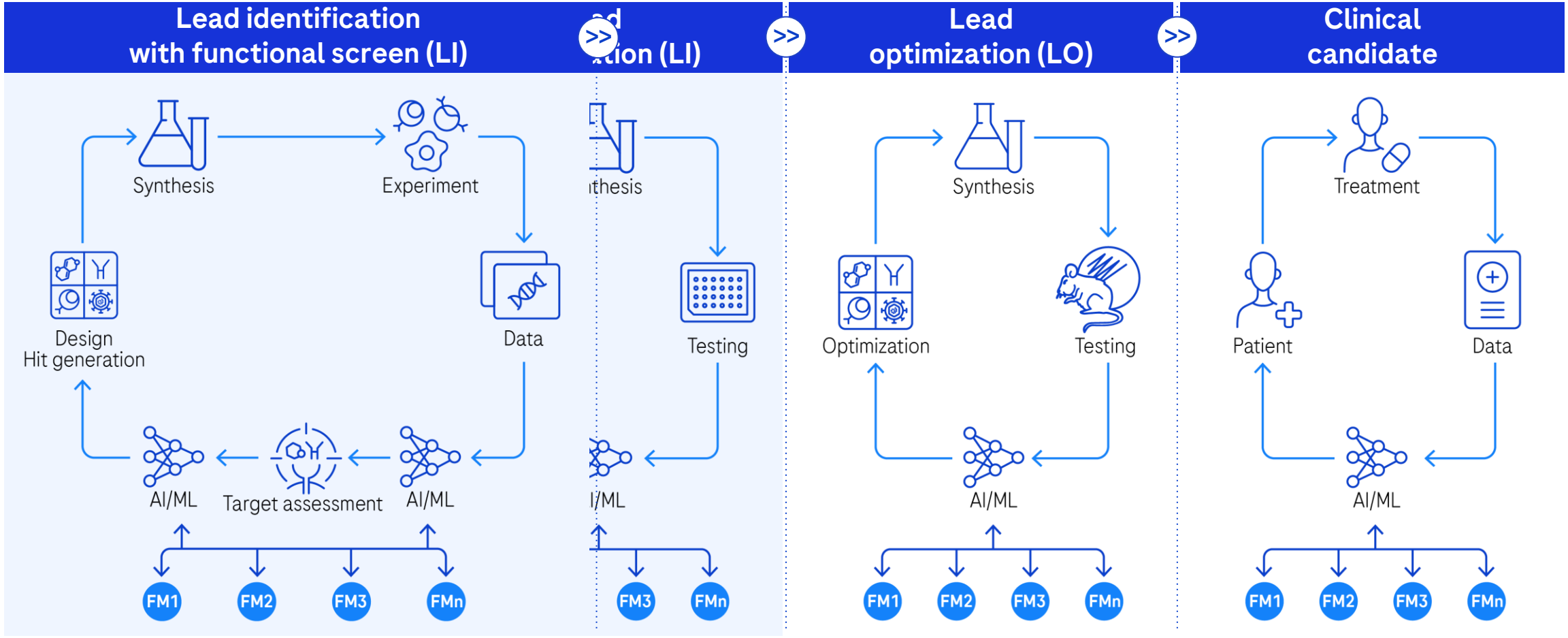
- Druggability
- Tractability
- Safety
- Modality

Clinical & Portfolio

- Patient population
- Biomarkers
- Portfolio fit
- Competitive landscape

Lab-in-the-Loop in drug R&D

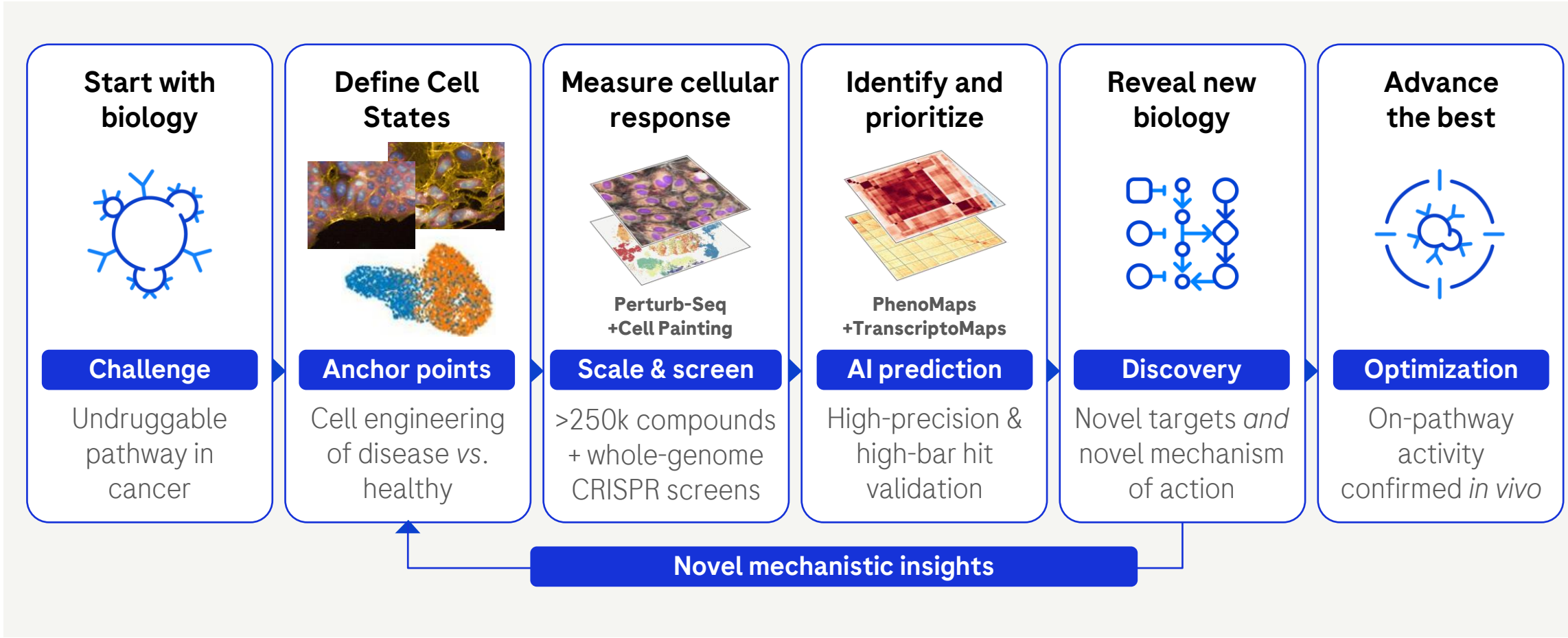
Iterations with AI embedded E2E, across each component, from target discovery to the clinic



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Lab-in-the-Loop applied to cancer target & drug discovery

Discovery of unique druggable MoA for a previously 'undruggable' pathway in oncology



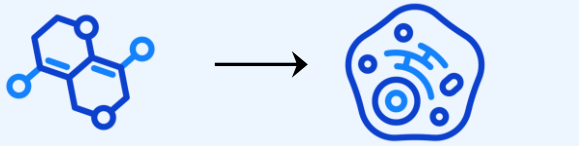
In collaboration with RECURSION

Virtual high-content small molecule phenotypic screens

Scale discovery from constrained lab process to dramatically larger virtual scale

~220K compounds screened in a lab, 90B (10^5-fold) space in ENAMINE to explore

Train foundation model

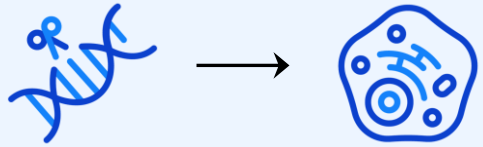


Input compound structure

Output measured phenotype

Example public dataset
116K compounds

Induce phenotype of interest in lab

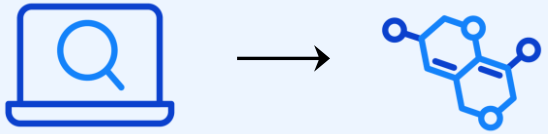


Anchor perturbation

Measure phenotype

Anchor perturbation

Conduct phenocopy virtual screen



Query the model virtually

Discover anchor phenocopies

Enamine virtual library
4B compounds screened

Prospective validation of a virtual cell phenotypic screen

mTOR/PI3K pathway inhibitors

Virtual screen and lab validation assays

Enamine REAL library

~4B compounds

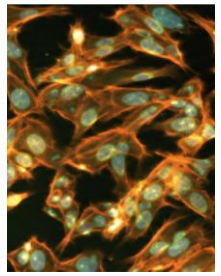
PhenoCompass

Predicted mTOR/PI3K pathway inhibitors

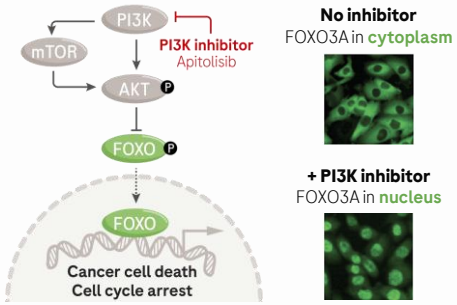
~500 compounds ordered (400 synthesized)

3 rounds of **experimental validation** with increasing specificity

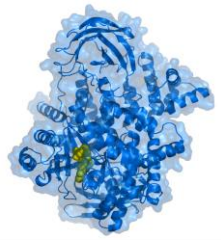
Morphology



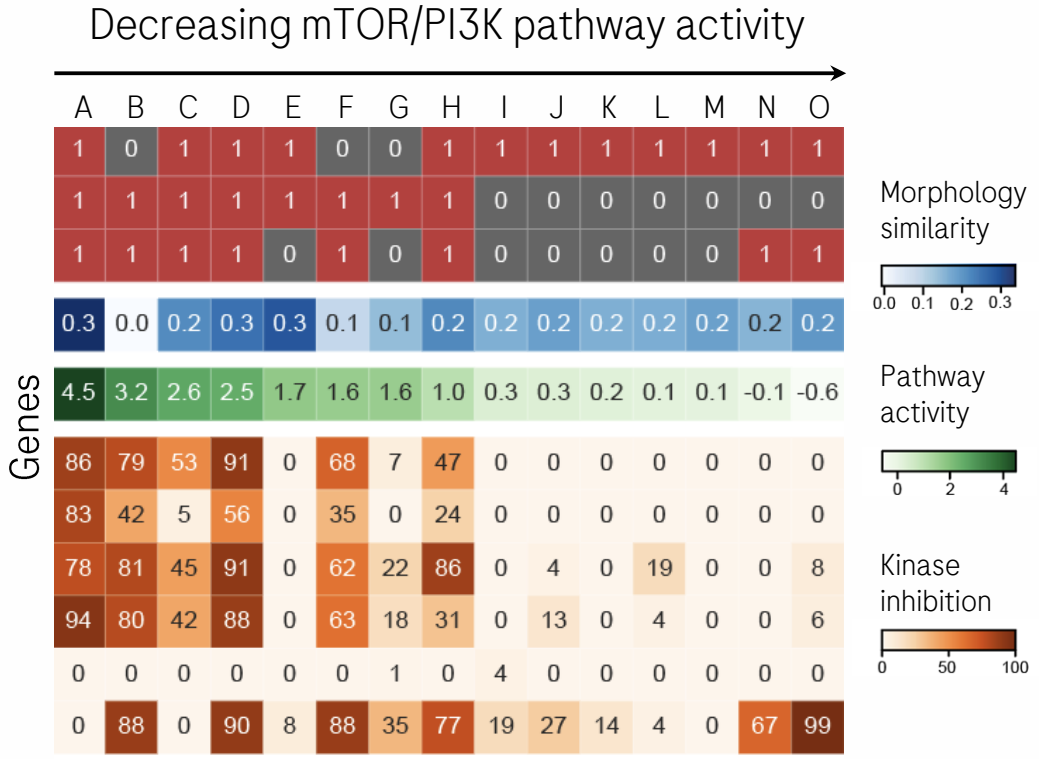
Pathway activity



Kinase inhibition

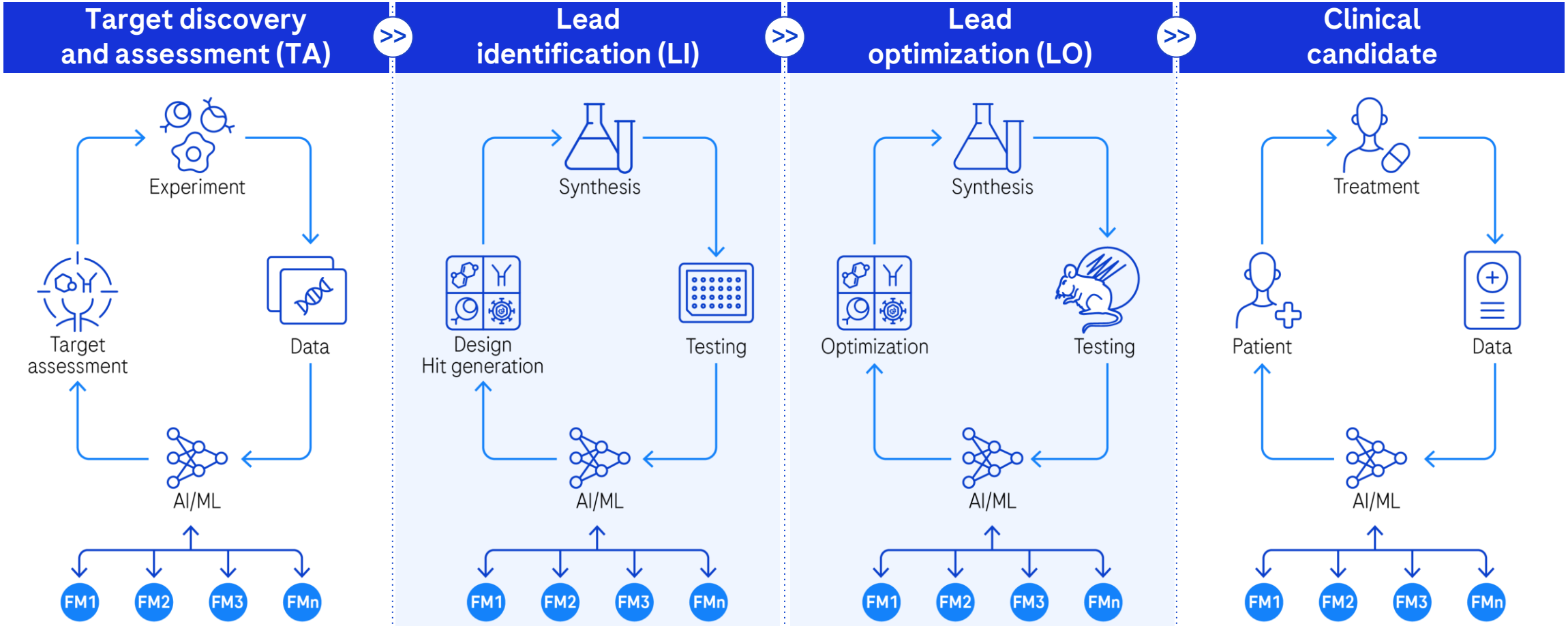


Hits with confirmed kinase inhibition; novel scaffolds



Lab-in-the-Loop in drug R&D

Iterations with AI embedded E2E, across each component, from target discovery to the clinic



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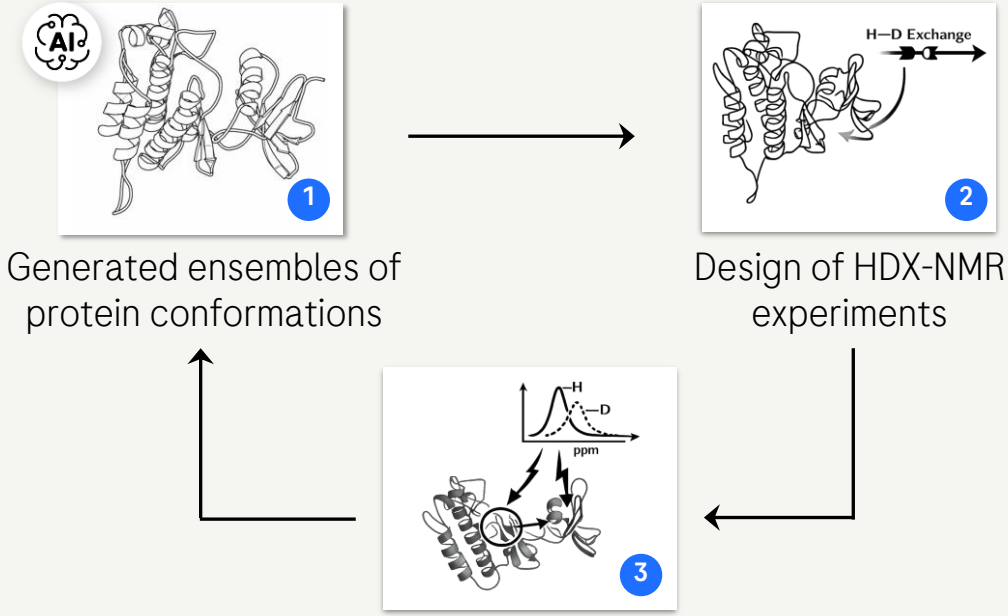
Small and large molecule drug discovery

AI can impact every stage of drug discovery

	Target assessment	Lead identification	Lead optimization
Small molecules	Druggability	Hit generation	Test, Analyze, Design, Make
Biologics	Lead discovery	Hit expansion	Test, Analyze, Design, Make

AI-powered structural biology accelerates hit finding

Identify druggable binding pockets



Generated ensembles of protein conformations

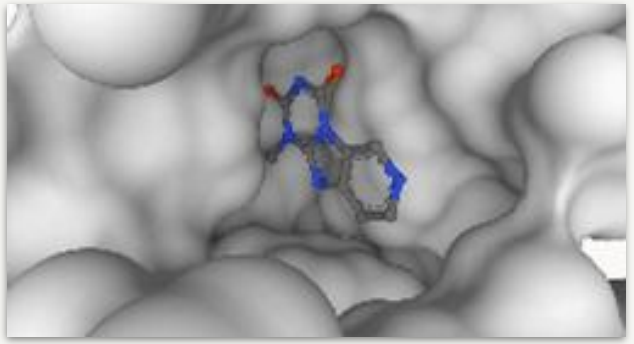
Design of HDX-NMR experiments

HDX-NMR maps solvent-exposed regions

Molecular hit finding

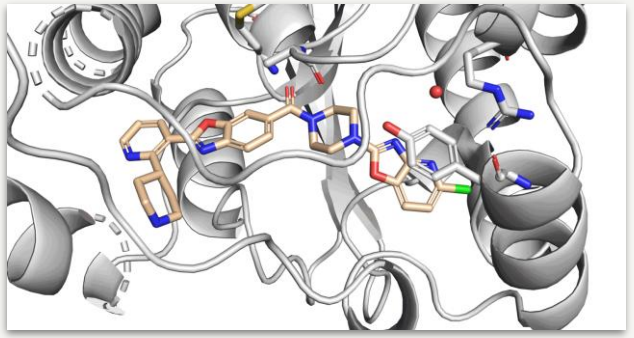
VoxBind

Pocket-conditioned voxel-denoising generative model



SAGE

Multi-property small molecule ligand design



Generic timeline — **X** months

First scaffold (New lab-based approach) — X **- 40%** months

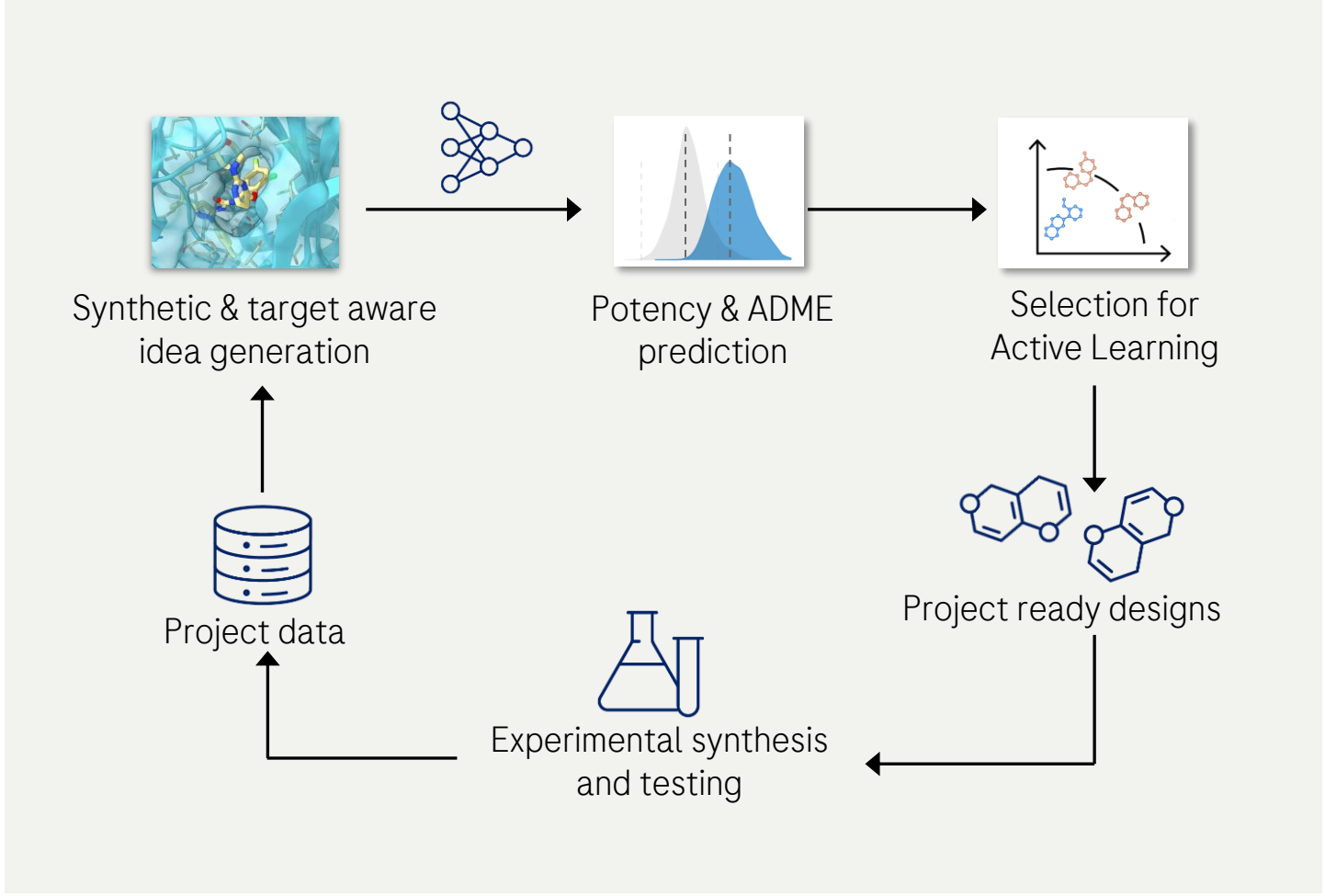
First scaffold (AI-enabled virtual screen) — X **- 80%** months — X **- 90%** months

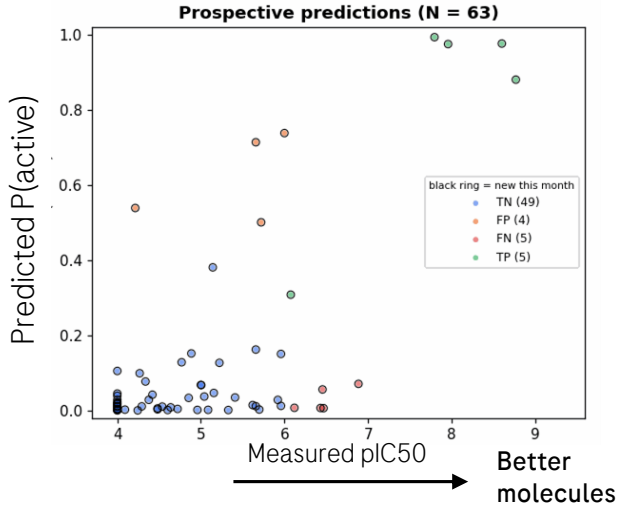
Hit #1 (first synthesis) — **Hit #2**

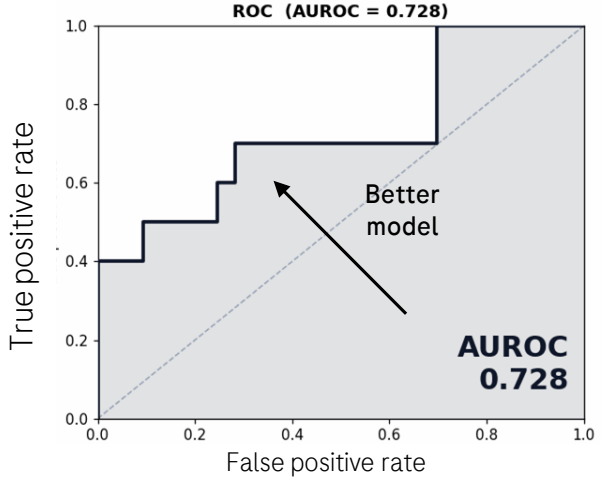
ICML 2026 GenBio Workshop Spotlight: Boltz-Jump: Accelerated Sampling of the Conformational Landscape of Biomolecular Structure Prediction Models. Daigavane *et al.*

Lab-in-the-Loop for small molecule optimization

Weekly active learning improves both molecules and models

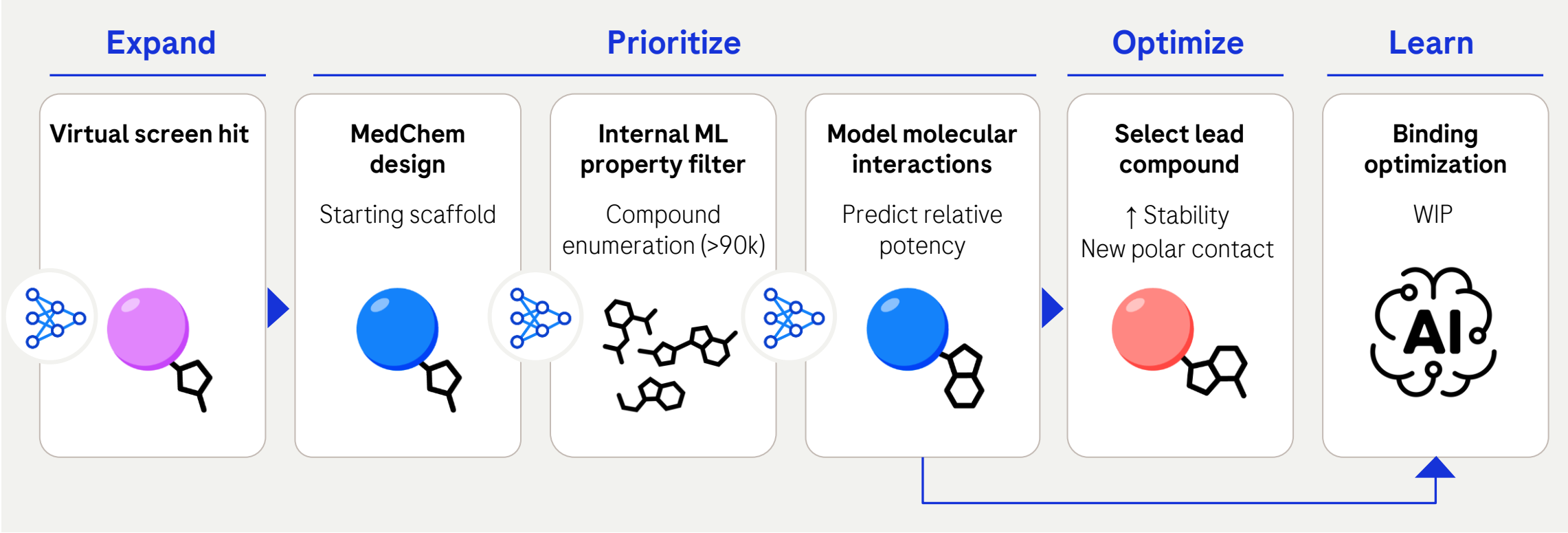






AI+physics-guided scaffold hopping accelerates LO

Optimizing metabolic stability while maintaining potency



Time
<35 min to generate & filter ~100K structures

Scale
>1k compounds

Potency
~40x improvement

An AI Agent supports SM drug discovery

Connecting context, SAR, design and computation to accelerate decisions

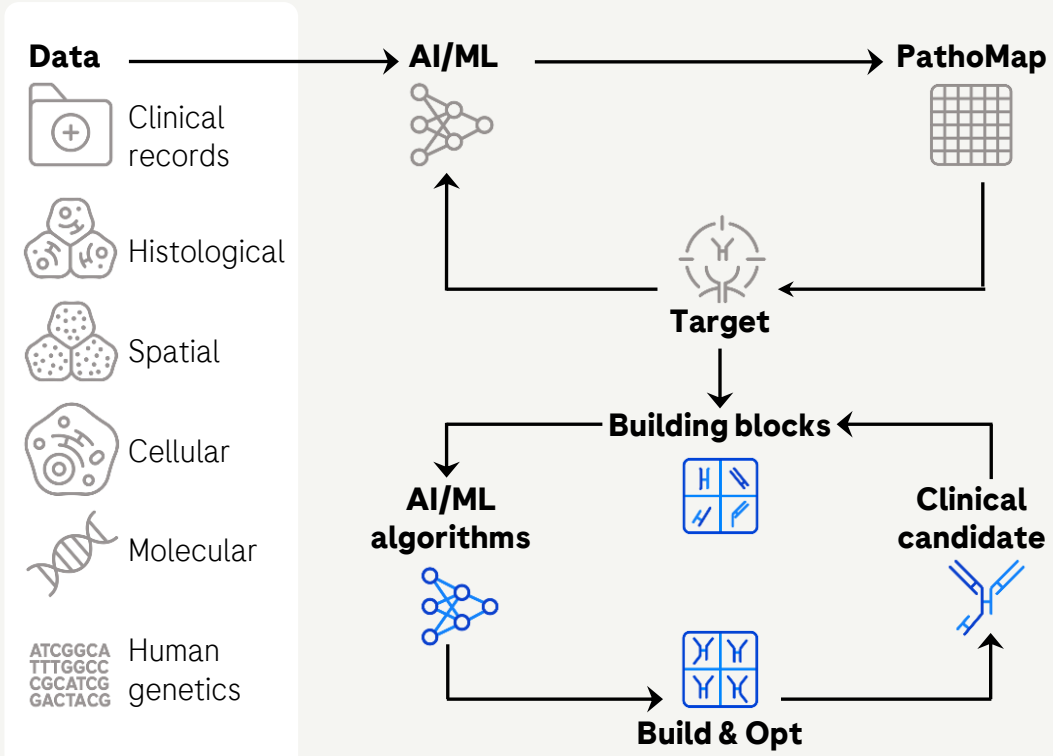
	Scientific foundation	AI support
Models & tools	Co-folding, docking, ADME and potency models, free energy perturbation, enumeration, etc.	**LLM + Harness**
Skills	Structural biology, structure-activity analysis, model evaluations, plotting and visualizations, etc.	**Reasoning** about project science and compound design
Data	Structure-activity project data, experimental structures, DesignHub, SM building blocks, etc.	**Orchestration** tools, models and workflows
Context	Project-specific context, documentations / history, medchem knowledge	

ADME, Absorption, distribution, metabolism, excretion; LLM, Large language model; SAR, Structure-activity relationship; SM, Small molecule

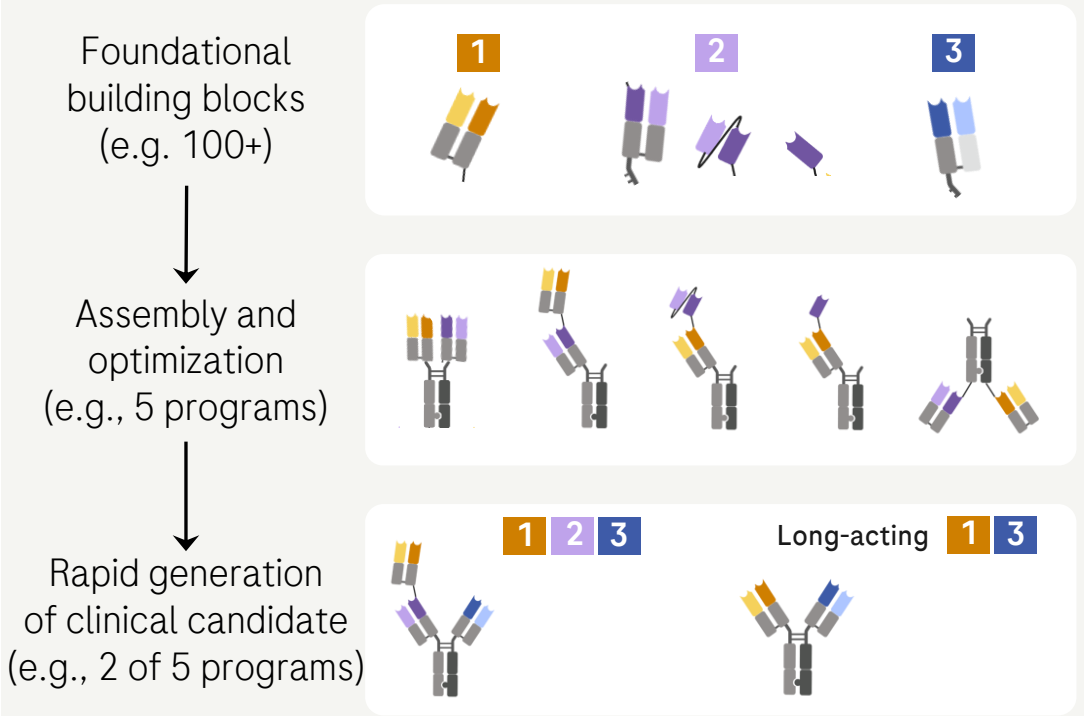
Building next-gen multispecifics to improve outcomes

Antibody building blocks and multi-program execution boost PTS, speed biologics, reduce costs

Data & AI driven target discovery



Foundational building blocks for next-gen multi-specifics against relevant targets



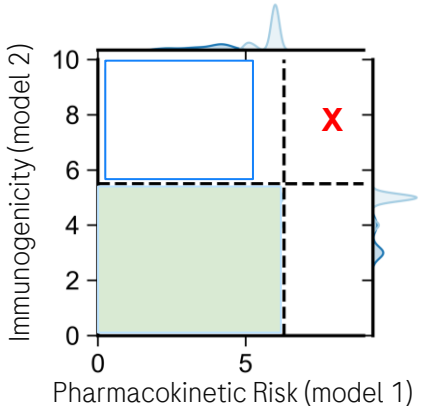
Integrated AI to optimize lead arms with low developability risk

Improved multi-specifics from building blocks

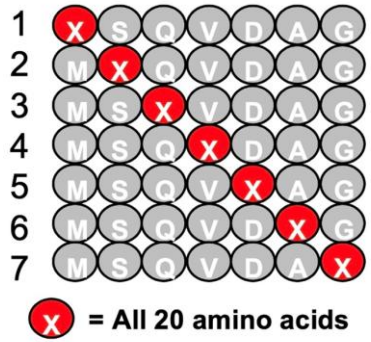
Example: Parallel efforts for each of 4 targets, for multiple bi- and tri-specifics

Rat, mice, rabbit, llama → High affinity and cross-reactivity → Potent function as monovalent arm → Lead and backup arms for each target

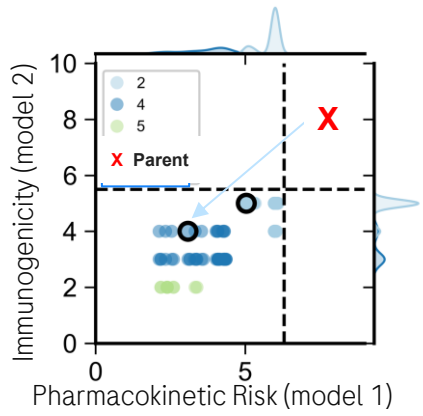
Integrated experiments and computation help optimize lead arms to lower developability risk

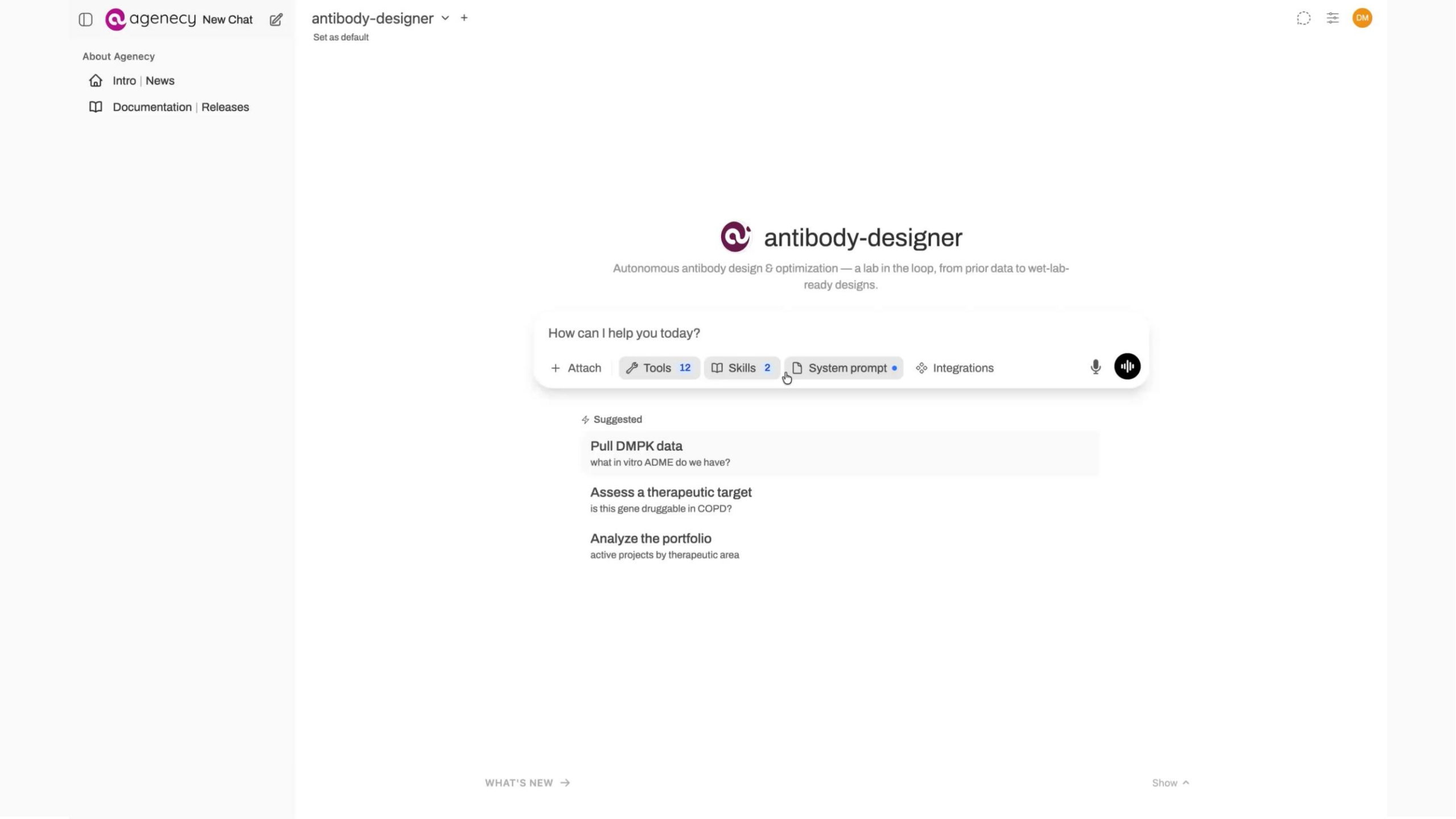


High-throughput experimental affinity data set: ~1,200 single mutants



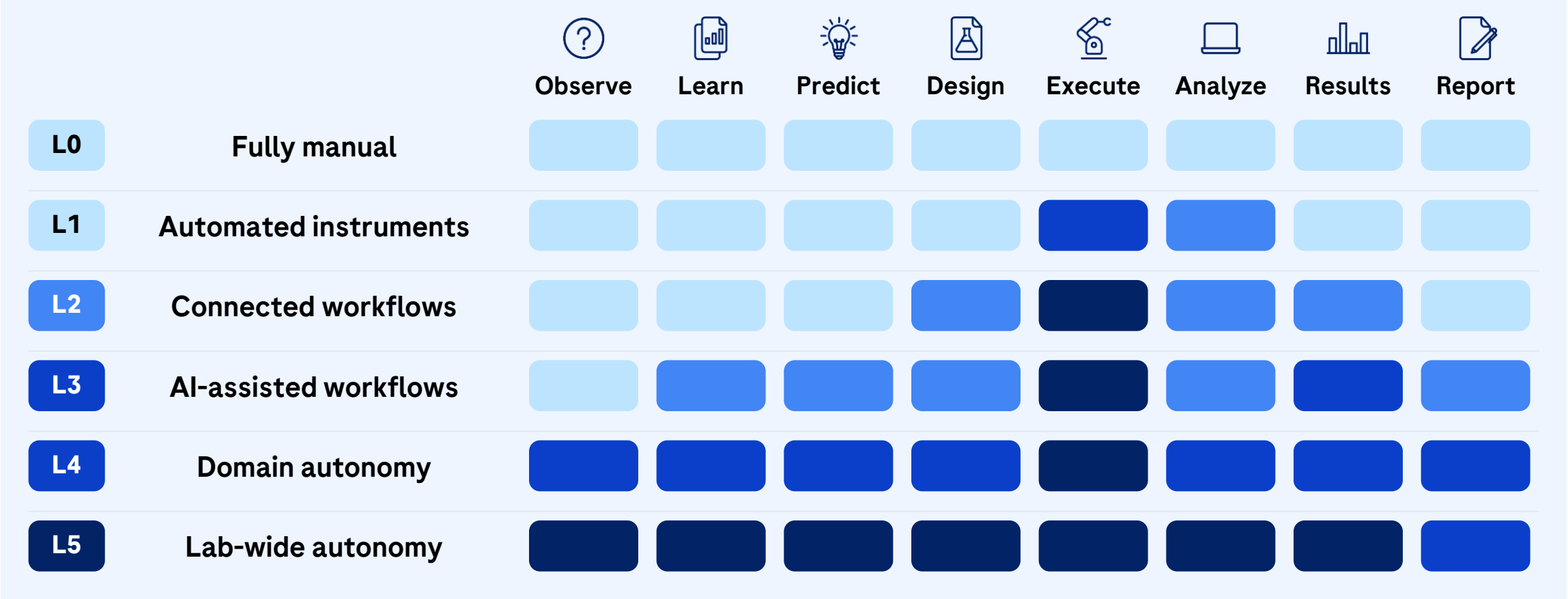
Sequence optimization




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antibody-designer
Autonomous antibody design & optimization — a lab in the loop, from prior data to wet-lab-ready designs.
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Tools 12
Skills 2
System prompt
Integrations
Suggested
Pull DMPK data
what in vitro ADME do we have?
Assess a therapeutic target
is this gene druggable in COPD?
Analyze the portfolio
active projects by therapeutic area
WHAT'S NEW
Show

Towards autonomous labs for drug R&D

Our Autonomous Lab Accelerator to enhance the loop



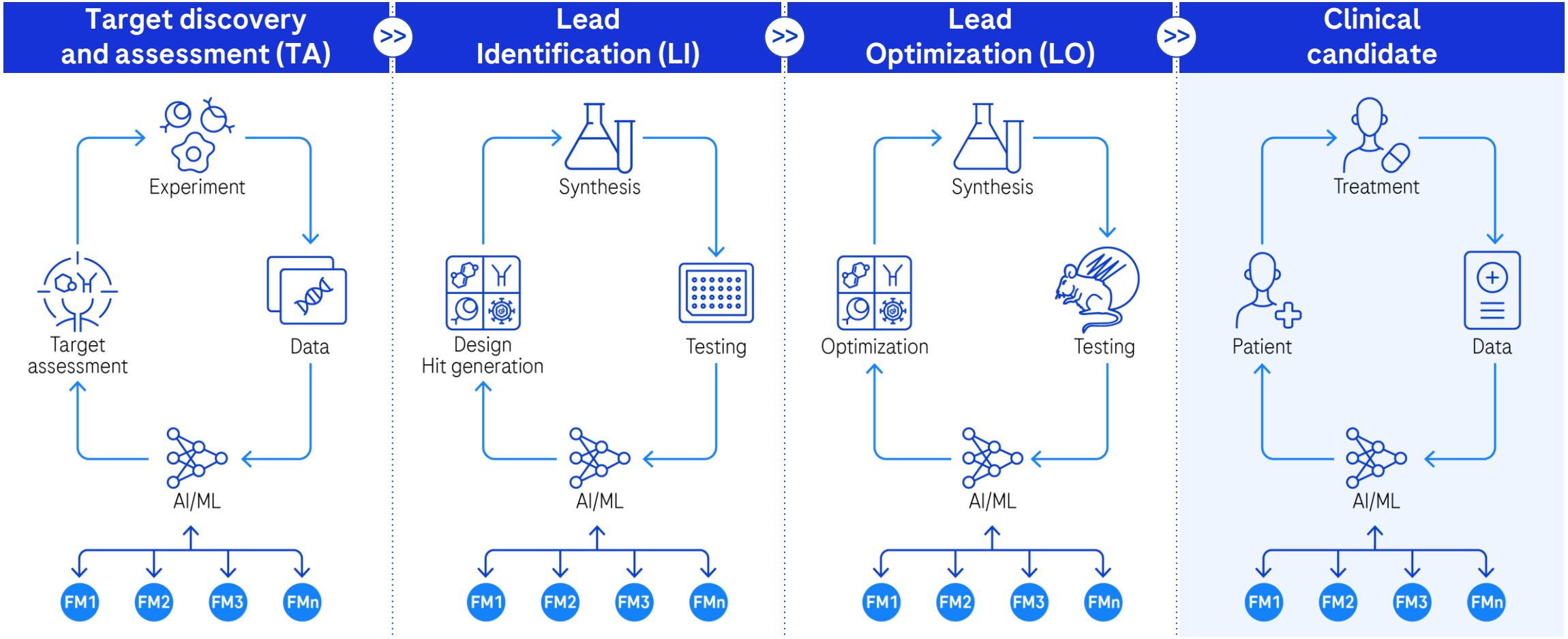
Human role: Operator | Assistant | Approver | Auditor

Including collaborations with Medra ANTHROP\C

Cheng et al. *Preprints.org* 2026 (in review)

Lab-in-the-Loop in drug R&D

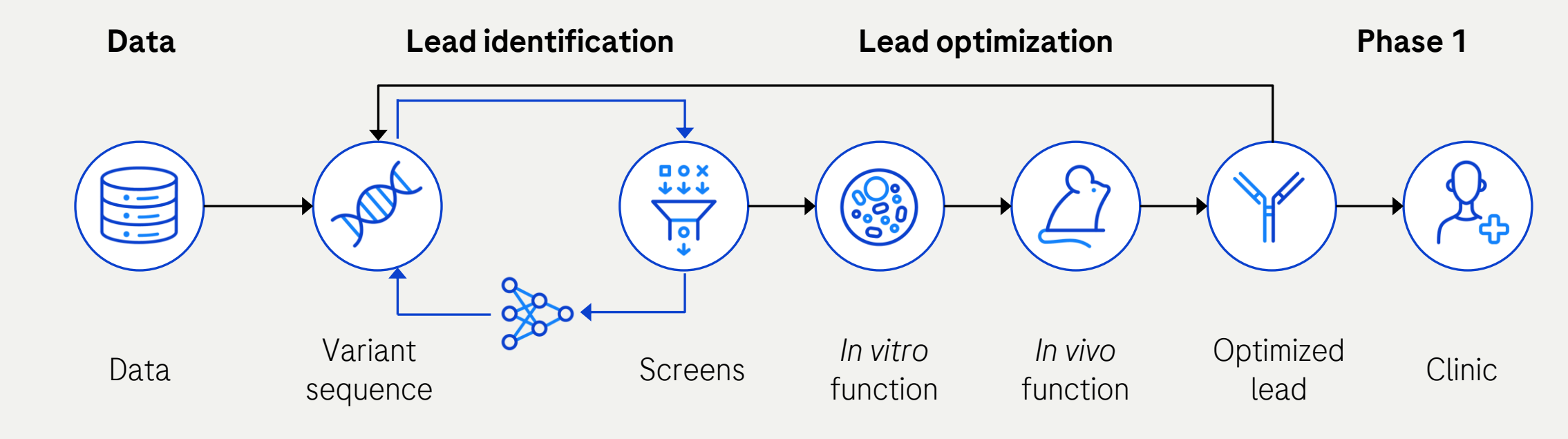
Iterations with AI embedded E2E, across each component, from target discovery to the clinic



AI/ML, Artificial Intelligence and Machine Learning; FM, Foundation model; LI, Lead identification; LO, Lead optimization; R&D, Research and development; TA, Target assessment

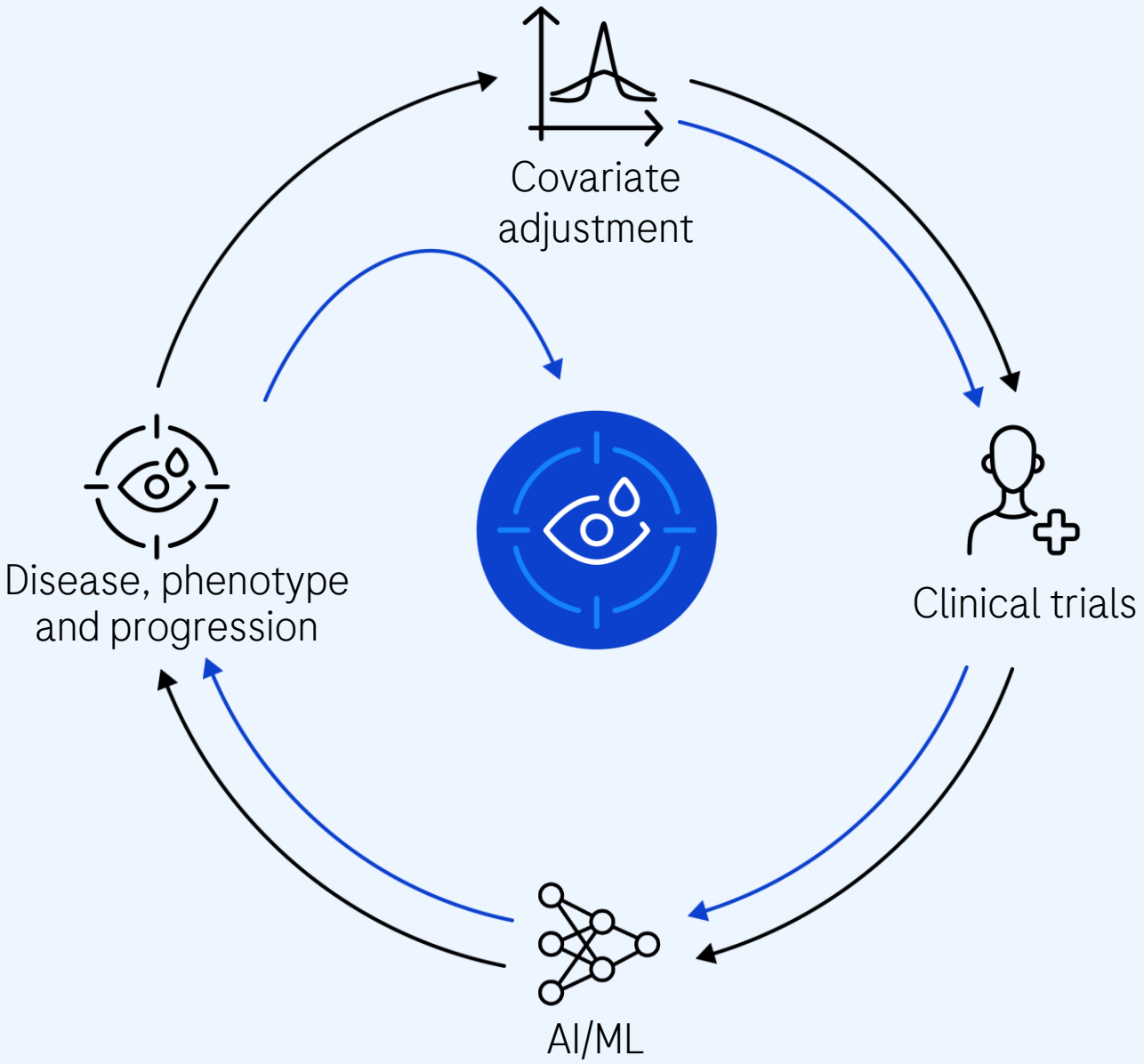
Lab-in-the-Loop helps discover better molecules, faster

Clinical-stage molecule with desired properties and reduced risk



LI to clinical candidate in 18 months: 2x faster, reduced immunogenicity risk

2022	2023	2024	2025	2026
Target identification (novel target)	Lead identification Go decision	Lead optimization Go decision	Clinical candidate	First patient in



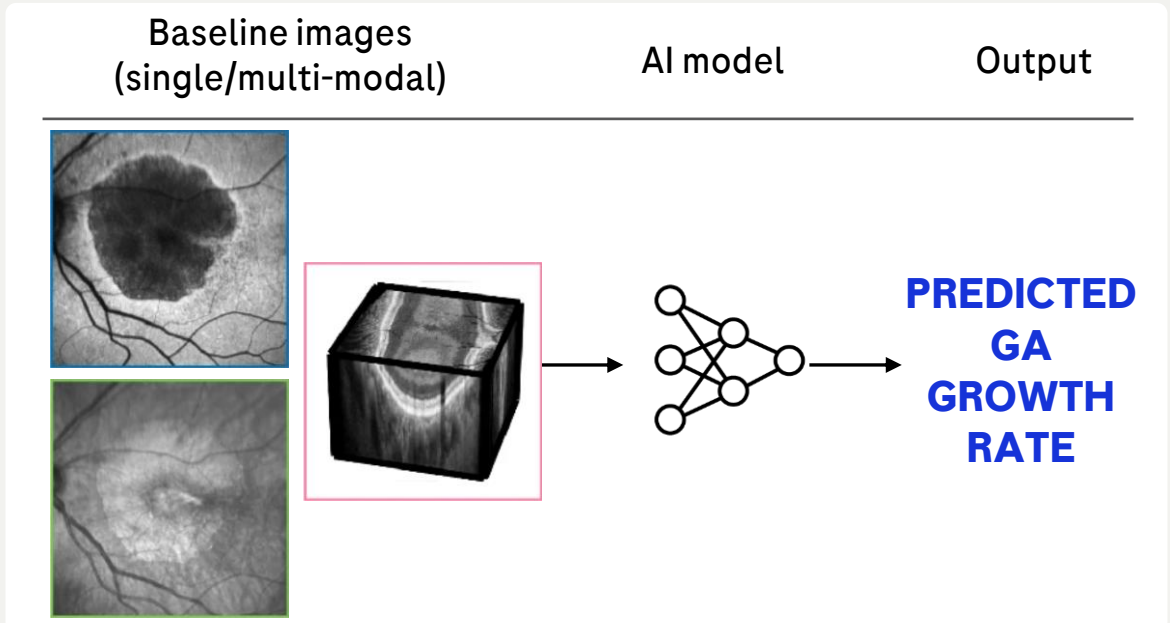
CLINIC IN THE LOOP

Lab-in-the-Loop to predict disease progression

AI outperforms in predicting lesion growth

AI-based models allows covariate adjustment and better clinical decision making

"E2E" AI to predict lesion growth rate

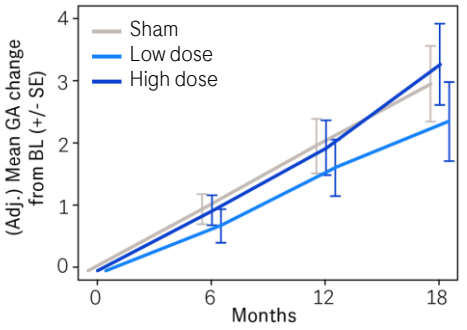


Trained and validated on >2,000 images from our past clinical trials in GA

Covariate adjustment with AI model

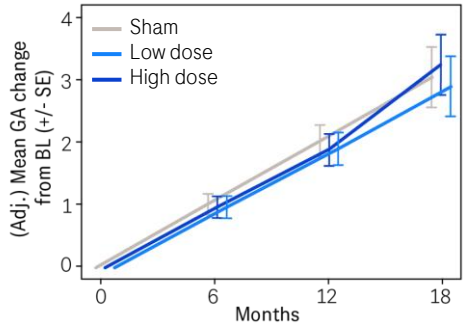
Mahalo **unadjusted** analysis

False-positive effect
~20% (p=0.16)



Mahalo **adjusted** analysis using deep-learning FAF model

No effect
~4.5% (p=0.6)



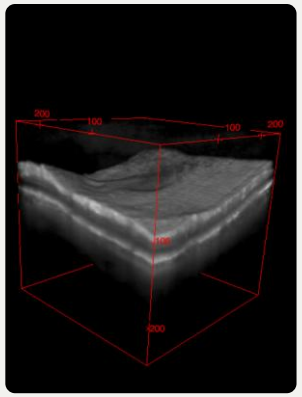
- Covariate Adjustment with AI model (which wasn't available at that time) changes Ph2 trial interpretation
- Precision of treatment effect estimate after adjustment is like running a trial >2x the size in this example

Anegondi N et al. *Ophthalmology Retina*. 2023 Mar;7(3):243-252.
E2E, End-to-end; FAF, Fundus autofluorescence; GA, Geographic atrophy; Ph2, Phase 2

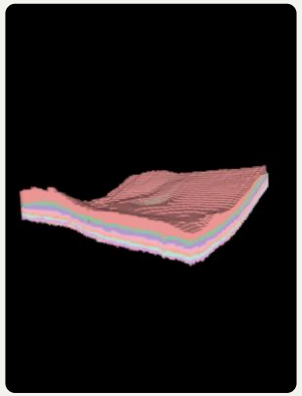
Clinic-in-the-Loop

GWAS of novel AI-based outcomes identifies loci not detected in traditional risk studies

OCT image

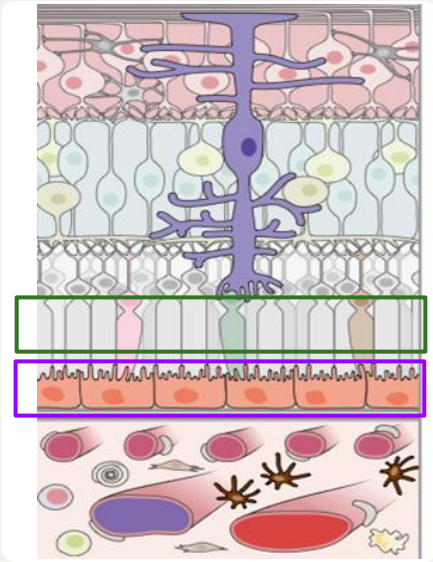


AI layer segmentation

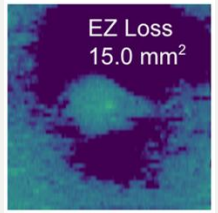


Ellipsoid zone (EZ):
photoreceptor health
RPE supports the photoreceptors

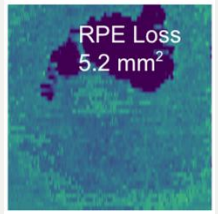
Retinal zone mapping



EZ loss
15.0 mm^2



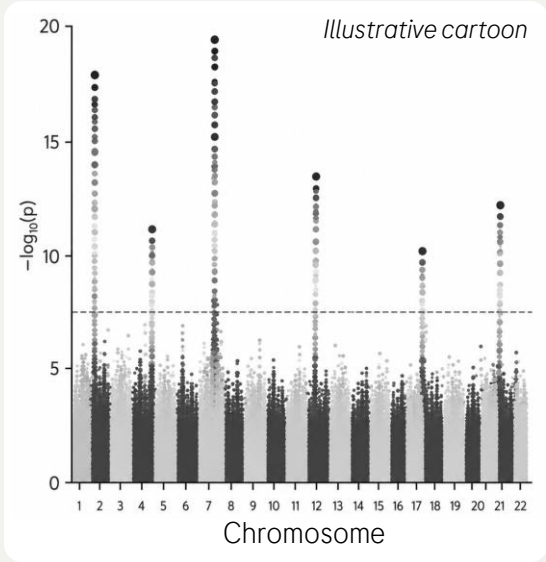
RPE loss
5.2 mm^2



$= \frac{15.0}{5.2}$

High **EZ**/**RPE**
= Poor patient prognosis

GWAS of novel outcomes

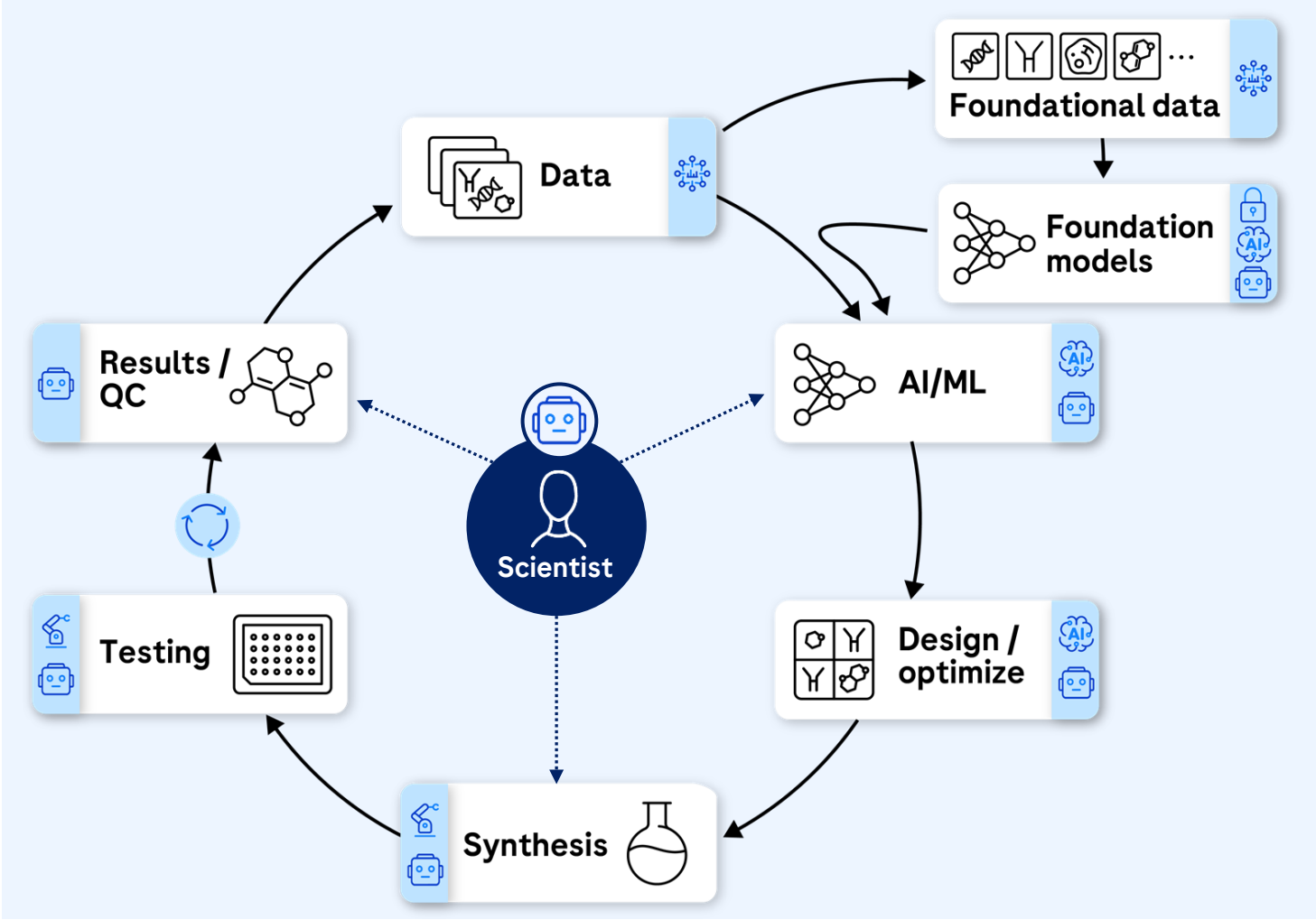


Closing the loop E2E: added as new input for GA pathobiology map driving **target discovery**

E2E, End-to-end; EZ, Ellipsoid zone; GWAS, Genome-wide association study; RPE, Retinal pigment epithelial cells; OCT, Optical coherence tomography

AI independence: owning the loop, learnings, improvements

Data generation and AI independence needed to maximize competitive benefit of our learnings



Capabilities

- Proprietary Foundation models
- Expert domain agents and harness
- AI factory
- Multiple data generation streams at scale
- Ability to iterate (lab and AL/RL)
- Autonomous lab capabilities

Partnerships

NVIDIA

RECURSION

Medra

ANTHROP\C